

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Worked Solutions

Advanced Sequences

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Solution

1. The first differences are 3, 7, 11, 15, 19,
2. Hence $b_k = 4k - 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 3$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 3$$

Worked Solutions

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: C

Final answer

$$a_n = n^2$$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: B

Final answer

$$a_n = \frac{3n^2 - 7n + 8}{2}$$

Worked Solutions

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.

Correct option: A

Final answer

$$a_n = 3n^2 - 8n + 12$$

Worked Solutions

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :**Solution**

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_5 = 25$

Correct option: B**Final answer**

25

Worked Solutions

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.
5. $a_6 = 37$

Correct option: A**Final answer**

37

Worked Solutions

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.
5. $a_6 = 72$

Correct option: D

Final answer

72

Worked Solutions

Advanced Sequences

Q008 Written

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.

Solution

1. The first differences are 8, 14, 20, 26, 32,
2. Hence $b_k = 2(3k + 1)$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (2(3k + 1)) = n(3n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = n(3n - 1)$$

Worked Solutions

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Solution

1. The first differences are 5, 9, 13, 17, 21,
2. Hence $b_k = 4k + 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (4k + 1) = n(2n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = n(2n - 1)$$

Worked Solutions

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Solution

1. The first differences are represented by $b_k = 5k + 4$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (5k + 4) = \frac{(n+1)(5n-2)}{2}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(5n-2)}{2}$$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Solution

1. The first differences are represented by $b_k = 7k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (7k - 3) = \frac{7n^2 - 13n + 14}{2}$.
4. Substituting $n = 1$ gives $a_1 = 4$, matching the given first term.

Correct option: D

Final answer

$$a_n = \frac{7n^2 - 13n + 14}{2}$$

Worked Solutions

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2(2k + 3)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 6 + \sum_{k=1}^{n-1} (2(2k + 3)) = 2n(n + 2)$.
4. Substituting $n = 1$ gives $a_1 = 6$, matching the given first term.

Correct option: B

Final answer

$$a_n = 2n(n + 2)$$

Worked Solutions

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Solution

1. The first differences are $3, 1, -1, -3, -5, \dots$
2. Hence $b_k = 5 - 2k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (5 - 2k) = -n(n - 6)$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -n(n - 6)$$

Worked Solutions

Advanced Sequences

Q014 Written

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.

Solution

1. The first differences are 1, 3, 9, 27, 81,
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 9}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 9}{6}$$

Worked Solutions

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Solution

1. The first differences are $1, 3, 9, 27, 81, \dots$
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 3}{6}$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 3}{6}$$

Worked Solutions

Advanced Sequences

Q016 Written

Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.

Solution

1. The first differences are 2, 4, 8, 16,
2. Hence $b_k = 2^k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (2^k) = 2^n + 3$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = 2^n + 3$$

Worked Solutions

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Solution

1. The first differences are $1, -2, 4, -8, 16, \dots$
2. Hence $b_k = (-2)^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Q018 Written

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.

Solution

1. The first differences are $-1, 3, -9, 27, -81, \dots$
2. Hence $b_k = \frac{(-3)^k}{3}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} \left(\frac{(-3)^k}{3}\right) = -\frac{(-3)^n - 57}{12}$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -\frac{(-3)^n - 57}{12}$$

Worked Solutions

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Solution

1. The first differences are represented by $b_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (k^2) = \frac{(n+1)(2n^2-5n+6)}{6}$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(2n^2-5n+6)}{6}$$

Worked Solutions

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Solution

1. The first differences are represented by $b_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 1$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_{12} = 133$; this is greater than 130, so immediate surgery is required.

Final answer

$$a_n = n^2 - n + 1, \quad a_{12} = 133; \text{ surgery required}$$

Worked Solutions

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 164

Final answer

164

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 163

Final answer

163

Worked Solutions

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 45$ terms precede the group.
2. First position $= P_9 + 1 = 46$
3. First value $= 46$

Final answer

46

Q024 MCQ

Select the correct option: find the first term of group 3:

Solution

1. $P_2 = 3$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 4.

Correct option: B

Final answer

4

Worked Solutions

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Solution**

1. $P_5 = 25$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 78.

Correct option: C**Final answer**

78

Q026 MCQ

Select the correct option: find the first term of group 9:**Solution**

1. $P_8 = 36$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 37.

Correct option: D**Final answer**

37

Worked Solutions

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 30.

Correct option: B**Final answer**

30

Q028 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 21$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 22.

Correct option: C**Final answer**

22

Worked Solutions

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 246.

Correct option: D**Final answer**

246

Q030 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 20.

Correct option: A**Final answer**

20

Worked Solutions

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 27$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 110.

Correct option: B**Final answer**

110

Q032 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 164.

Correct option: C**Final answer**

164

Worked Solutions

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Solution**

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 58.

Correct option: A**Final answer**

58

Q034 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 49$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 100.

Correct option: B**Final answer**

100

Worked Solutions

Advanced Sequences

Q035 **Written**

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 126$ terms precede the group.
2. First position = $P_9 + 1 = 127$
3. First value = 379

Final answer

379

Q036 **MCQ**

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 29.

Correct option: D

Final answer

29

Worked Solutions

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 28$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 89.

Correct option: A**Final answer**

89

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .**Solution**

1. First value $= 2(n^2 - 2n + 2)$
2. Last value $= 2n^2$
3. Number of terms $= 2n - 1$
4. $S_n = \frac{c_n}{2}(\text{first} + \text{last}) = 2(2n - 1)(n^2 - n + 1)$

Final answer $2(2n - 1)(n^2 - n + 1)$

Worked Solutions

Advanced Sequences

Q039 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 163
2. Last value = 199
3. Number of terms = 19
4. $S_{10} = 3439$

Final answer

3439

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 46
2. Last value = 55
3. Number of terms = 10
4. $S_{10} = 505$

Final answer

505

Worked Solutions

Advanced Sequences

Q041 **Written****For the same grouped sequence, find the sum of the terms in group 10.****Solution**

1. First value = 379
2. Last value = 463
3. Number of terms = 29
4. $S_{10} = 12209$

Final answer

12209

Q042 **Written****In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.****Solution**

1. Sequence position = $(88 - 2)/2 + 1 = 44$
2. $P_6 = 36 < 44 \leq P_7 = 49$
3. Therefore the value is in Group 7, position 8.

Final answer

Group 7, position 8

Worked Solutions

Advanced Sequences

Q043 Written

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Solution

1. Sequence position $= (101 - 1)/2 + 1 = 51$
2. $P_7 = 49 < 51 \leq P_8 = 64$
3. Therefore the value is in Group 8, position 2.

Final answer

Group 8, position 2

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Solution

1. Sequence position $= (124 - 1)/1 + 1 = 124$
2. $P_{15} = 120 < 124 \leq P_{16} = 136$
3. Therefore the value is in Group 16, position 4.

Final answer

Group 16, position 4

Worked Solutions

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Solution

1. Sequence position $= (100 - 1)/3 + 1 = 34$
2. $P_4 = 26 < 34 \leq P_5 = 40$
3. Therefore the value is in Group 5, position 8.

Final answer

Group 5, position 8

Worked Solutions

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Solution

1. $c_n = n + 1, \quad P_n = \frac{n(n+3)}{2}$
2. (a) first term = 310
3. (b) group sum = $2(n+1)(n^2 + 2n - 1)$
4. (c) 202 is in Group 9, position 7.
5. (d) total = 16200

Final answer

(a) 310; (b) $2(n+1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Worked Solutions

Advanced Sequences

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 20}{6}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: C

Final answer

$$a_n = \frac{(-2)^n + 20}{6}$$

Worked Solutions

Advanced Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 64$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 195.

Correct option: A

Final answer

195

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Solution

1. The value is sequence position 15.
2. It lies in Group 5, position 1.

Correct option: B

Final answer

Group 5, position 1

Worked Solutions

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = 4k - 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 2$$

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Solution

1. $P_{11} = 66$ terms precede the group.
2. First position $= P_{11} + 1 = 67$
3. First value $= 203$

Final answer

203

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Solution

1. Sequence position $= (66 - 2)/4 + 1 = 17$
2. $P_4 = 14 < 17 \leq P_5 = 20$
3. Therefore the value is in Group 5, position 3.

Final answer

Group 5, position 3

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3